Controlled Class-Imbalance Construction for TCGA-UT

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Abstract

Class imbalance in computational pathology is not only a question of how many examples a class contributes. A class may perform poorly because it is rare, because its frozen foundation-model features overlap with other classes, or because both effects coincide. The companion TCGA-UT imbalance benchmark established the dataset, the mitigation taxonomy, and patch-level and WSI-bag evaluations under the native TCGA-UT long tail. This report defines a controlled counterpart: artificial full-scale TCGA-UT training distributions that keep all 32 cancer types, preserve fixed validation and test splits, and vary only the training support assigned to each class. Power-law training distributions are generated under native-prevalence class ordering at imbalance parameters $\lambda \in \{0.8, 1.1, 1.3\}$ with one shared strictly feasible pool of 1,528 slides, yielding realized head-to-tail ratios of approximately 15:1, 43:1, and 78:1 that bracket the native TCGA-UT long tail. Using frozen Virchow2 patch features and WSI-bag features with the same 30-instance evidence budget as the companion benchmark, the controlled study is configured to mirror the first report's method families under exact-feasibility sampling rather than redistribution-based fallback. Method rankings are broadly preserved across imbalance severities, while balanced accuracy declines monotonically by three to five percentage points from $\lambda = 0.8$ to $\lambda = 1.3$.

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1 Introduction

The companion TCGA-UT imbalance benchmark compared class-imbalance mitigation methods in their native input regimes: labeled patch-feature classification and weakly supervised WSI-bag classification. It also showed that class frequency alone is an incomplete explanation for poor classwise performance. Some low-support cancer types remain separable under a frozen pathology foundation model, whereas other classes are difficult even when their support is not the smallest. This report turns that observation into an experimental design question: how should mitigation methods be interpreted when class frequency and inherent class difficulty can be varied deliberately?

TCGA-UT is a useful test bed for this question because it provides both slide labels and patch-level evidence derived from diagnostic whole-slide images in The Cancer Genome Atlas (Tomczak et al., 2015; Komura et al., 2022). The native dataset already has a long-tailed slide distribution over 32 cancer types. Native imbalance is realistic, but it entangles prevalence with many other factors: tumor morphology, tissue diversity, slide acquisition, and the representation geometry induced by the frozen encoder. A controlled construction complements the native benchmark by imposing known training-support profiles while holding validation and test data fixed.

The goal is not to introduce a new mitigation method. Instead, the report evaluates the same method families as the companion taxonomy under three constructed training distributions that vary only the imbalance severity while holding the class ranking fixed. Patch-level methods include cross-entropy, class weighting, balanced sampling, focal loss, Scholz-style metric losses, CFAL, ProGAN augmentation, and OKO set learning (Scholz et al., 2024). WSI-bag methods include cross-entropy, class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Dataset Construction

The construction starts from the same TCGA-UT case-disjoint split protocol as the companion benchmark. For each seed, the validation and test splits are kept fixed, and artificial imbalance is applied only to the training split. The split assignment is patient-disjoint: all slides and feature rows associated with one TCGA participant remain in exactly one of train, validation, or test. This isolates the effect of training support: model selection and held-out evaluation remain comparable across imbalance parameters and mitigation methods without patient leakage.

2.1 Full-Scale Training Pool

All 32 TCGA-UT cancer types are retained. Let $C = 32$ and let $\mathcal{D}_{\text{train}}$ denote the seed-specific training slide pool. For class c, let a_c be the number of available training slides before artificial resampling. The constructed training split must satisfy three constraints:

$$1 \leq n_c \leq a_c, \qquad \sum_{c=1}^{C} n_c = N, \qquad c \in \{1, \ldots, C\}. \tag{1}$$

The first constraint keeps every class represented and prevents oversampling beyond available slides. The second fixes the training pool at N slides, shared across all imbalance parameters and seeds; the choice of N is discussed in Section 2.3. The third makes the benchmark a 32-class TCGA-UT study throughout. Unlike the earlier prototype code path, the active benchmark does not redistribute overflow mass or sample with replacement: an infeasible target distribution is treated as an invalid experimental request.

After slide selection, the input evidence is capped in the same way as in the companion benchmark. Patch classifiers use up to 30 deterministic patches per slide. WSI-bag classifiers use up to 30 deterministic frozen feature instances per slide. In both regimes, Virchow2 features remain frozen (Vorontsov et al., 2024; Zimmermann et al., 2024). Features are read from the shared cls_patchmean Virchow2 store ($\mathrm{concat}(\mathrm{CLS}, \mathrm{mean}(\text{patch tokens}))$, 2,560-dimensional vectors) used by the companion WSI-bag benchmark. The cap prevents slides with more extracted tumor patches from dominating the training objective while keeping the input budget aligned across regimes.

2.2 Power-Law Training Support

For an ordered class list $(c_1, \ldots, c_C)$, the intended share for rank i under imbalance parameter λ is

$$p_i(\lambda) = \frac{i^{-\lambda}}{\sum_{j=1}^{C} j^{-\lambda}}. \tag{2}$$

When $\lambda = 0$, the intended distribution is uniform over ranks. As λ increases, early-ranked classes receive more training support and late-ranked classes receive less. Raw targets $|\mathcal{D}_{\text{train}}|p_i(\lambda)$ are converted to integer counts by flooring and assigning residual slides by largest fractional remainder.

Each class first receives one guaranteed slide. The remaining $N - C$ slides are then allocated according to Equation (2), again using largest-remainder rounding to obtain integer counts. A requested regime is accepted only when every rounded class target satisfies Equation (1). If any class target exceeds its available training support, the construction fails and the pool size must be reduced; there is no redistribution or replacement fallback.

2.3 Class Order and Training Pool

All constructed splits use the *native-prevalence order*: classes are ranked by their TCGA-UT training-split slide count, so that rank 1 is the most frequent cancer type and rank $C = 32$ is the least frequent. This ordering follows the realistic direction of the native TCGA-UT long tail.

Pool size. The choice of pool size N determines whether the requested regimes are feasible at all. When $N = \sum_c a_c$ (the full available training pool), the steepest power-law targets would exceed the capacity of the rare classes, so the benchmark would have to rely on a hidden fallback mechanism. To keep the construction exact, N must instead be small enough that no class target exceeds its available support even at the steepest imbalance.

Define the maximum clean pool for a given λ and seed as

$$T^*(\lambda) = \min_c \frac{a_c}{p_c(\lambda)}. \tag{3}$$

For $N \le T^*(\lambda)$, the intended targets $N\,p_c(\lambda)$ are feasible for every class without fallback. Setting $N = \lfloor \min_s T_s^*(1.3) \rfloor$, where the outer minimum is over the three seeds, guarantees that the steepest evaluated imbalance produces an exact feasible power-law distribution on every seed. For the TCGA-UT training splits, this yields $N =$ 1,528 slides, approximately 25% of the full training pool.

Realized support profiles. The three λ values produce head-to-tail ratios of approximately 15:1 ($\lambda = 0.8$), 43:1 ($\lambda = 1.1$), and 78:1 ($\lambda = 1.3$), bracketing the native TCGA-UT ratio of approximately 28:1. The $\lambda = 0.8$ setting places the constructed distribution below native imbalance; $\lambda = 1.1$ is moderately more severe; and $\lambda = 1.3$ is the strongest regime evaluated.

These three operating points verify whether mitigation methods provide useful protection as imbalance intensifies along a common class ranking.

3 Methods

The method taxonomy follows the companion benchmark, but this report uses it to study controlled training distributions rather than only the native TCGA-UT distribution. Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or synthetic augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, prototype-affinity objectives, or OKO set supervision. WSI-specific methods change how slide bags are represented or mixed.

Regime	Family	Methods	Controlled question
Patch	Baseline and losses	CE, weighted CE, focal loss	Frequency weighting and hard-example emphasis.
Patch	Metric losses	CE + soft F1, CE + soft MCC	Macro-metric losses with balanced minibatch exposure.
Patch	Specialized	CFAL, OKO	Prototype-affinity and set-level supervision.
Patch	Synthetic augmentation	ProGAN augmentation	Tail-class synthetic patch generation embedded through the frozen encoder.
WSI bag	Baseline and losses	CE, weighted CE, focal loss	Slide-level analogues of standard losses; weight_power=0 recovers plain MIL CE.
WSI bag	MIL-specific	Balanced sampling, RankMix, SC-MIL, MDE-MIL	Bag-level sampling, augmentation, contrastive structure, and expert training.

Table 1: Method groups evaluated under constructed TCGA-UT training distributions. Patch and WSI-bag methods are compared only within their own input regime.

Let y be the class label, $\mathbf{z} \in \mathbb{R}^C$ the logits, and p_y the softmax probability of the true class. Cross-entropy optimizes $-\log p_y$. Weighted cross-entropy multiplies this term by a normalized inverse-frequency weight. Focal loss uses

$$\mathcal{L}_{\text{focal}}(\mathbf{z}, y) = -(1 - p_y)^\gamma \log p_y, \tag{4}$$

so confidently classified examples contribute less to the gradient. The soft-F1 and soft-MCC variants add differentiable approximations of macro evaluation metrics to CE, following the imbalance-aware medical-image objectives of Scholz et al. (2024). RankMix, SC-MIL, and MDE-MIL are retained as WSI-bag methods because their mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025).

4 Evaluation Protocol

Each constructed split is indexed by seed and imbalance parameter $\lambda \in \{0.8, 1.1, 1.3\}$. For every method and constructed regime, hyperparameters are selected by validation macro F1 across three seeds; ties are broken by validation balanced accuracy. Selected configurations are evaluated on the fixed test split, and results are reported as mean ± standard deviation across seeds. Macro F1 is the primary discrimination metric because it weights cancer types equally. Balanced accuracy and accuracy are reported as complementary discrimination metrics.

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the case-disjoint splits, frozen Virchow2 representation family, and 30-instance evidence budget, but they differ in prediction unit and aggregation mechanism.

Calibration is evaluated alongside discrimination using negative log likelihood (NLL), multiclass Brier score (Brier, 1950), and expected calibration error (ECE; Naeini et al., 2015; Guo et al., 2017); full metric definitions and the temperature-scaling rationale follow the companion class-imbalance benchmark. For each method and seed, a single temperature is fit by minimizing NLL on the validation split and applied unchanged to the test score matrix before softmax (Guo et al., 2017). Tables 5 and 6 report both raw and temperature-scaled scores for each constructed regime.

5 Results

The result analysis asks how methods respond as imbalance severity increases from $\lambda = 0.8$ to $\lambda = 1.3$ under the native-prevalence class order. Table 2 summarises the realized support profiles; Figure 1 shows their per-rank structure. Tables 3 and 4 compare mitigation methods within the patch-feature and WSI-bag regimes.

λ	Training slides	Support range	Head:tail ratio
0.8	1,528	18–268	≈15:1
1.1	1,528	10–425	≈43:1
1.3	1,528	7–543	≈78:1
native (full pool)	6,091	20–562	≈28:1

Table 2: Constructed training-support profiles under native-prevalence ordering. Support range and head:tail ratio are the minimum and maximum per-class slide counts averaged across seeds. The native full-pool row is shown for reference; it is not part of the constructed experiment.

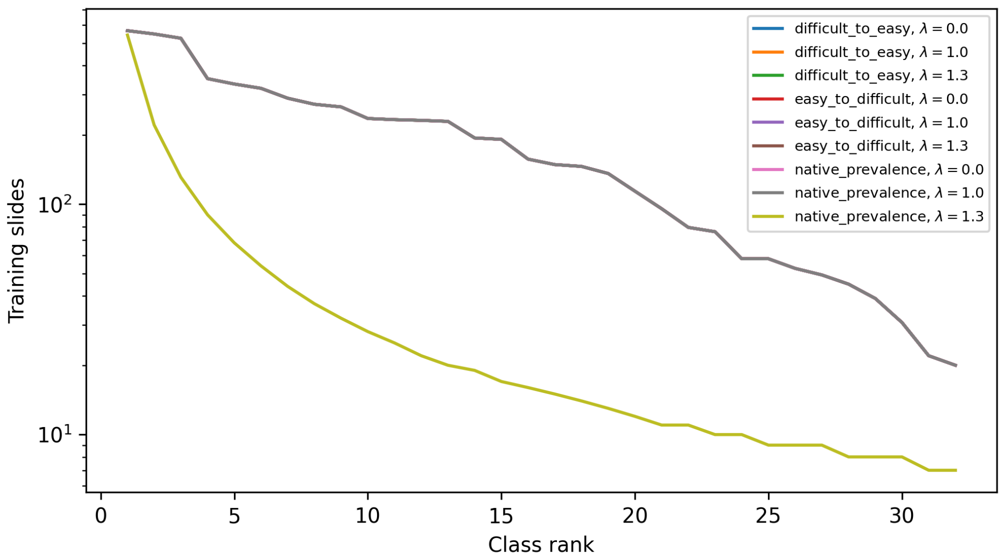

Figure 1: Constructed training support by class rank for the main imbalance settings.

Patch-feature results show a consistent, monotone decline as imbalance intensifies (Table 3). At $\lambda = 0.8$, CFAL, OKO, and ProGAN augmentation are tied for the highest balanced accuracy at 0.723; focal loss and weighted CE follow at 0.718. At $\lambda = 1.1$, OKO leads at 0.707, ahead of CFAL and CE+soft-F1 (both 0.703). At $\lambda = 1.3$, CE+soft-F1 emerges with the highest

Method	Accuracy	Balanced accuracy	Macro F1
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 0.8$)	0.763 ± 0.013	0.712 ± 0.004	0.693 ± 0.006
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 1.1$)	0.761 ± 0.020	0.694 ± 0.021	0.686 ± 0.027
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 1.3$)	0.746 ± 0.014	0.672 ± 0.009	0.671 ± 0.009
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 0.8$)	0.759 ± 0.009	0.718 ± 0.005	0.685 ± 0.002
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 1.1$)	0.752 ± 0.015	0.703 ± 0.020	0.680 ± 0.018
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 1.3$)	0.747 ± 0.016	0.688 ± 0.007	0.668 ± 0.008
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 0.8$)	0.757 ± 0.009	0.709 ± 0.013	0.690 ± 0.007
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 1.1$)	0.758 ± 0.014	0.692 ± 0.014	0.681 ± 0.020
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 1.3$)	0.739 ± 0.013	0.668 ± 0.014	0.658 ± 0.012
patch_feature_cfal (native_prevalence, $\lambda = 0.8$)	0.774 ± 0.011	0.723 ± 0.004	0.699 ± 0.007
patch_feature_cfal (native_prevalence, $\lambda = 1.1$)	0.776 ± 0.016	0.703 ± 0.013	0.705 ± 0.015
patch_feature_cfal (native_prevalence, $\lambda = 1.3$)	0.762 ± 0.013	0.681 ± 0.010	0.685 ± 0.013
patch_feature_focal (native_prevalence, $\lambda = 0.8$)	0.769 ± 0.010	0.718 ± 0.004	0.698 ± 0.004
patch_feature_focal (native_prevalence, $\lambda = 1.1$)	0.768 ± 0.016	0.696 ± 0.015	0.690 ± 0.018
patch_feature_focal (native_prevalence, $\lambda = 1.3$)	0.754 ± 0.010	0.672 ± 0.006	0.670 ± 0.013
patch_feature_oko (native_prevalence, $\lambda = 0.8$)	0.769 ± 0.020	0.723 ± 0.014	0.695 ± 0.020
patch_feature_oko (native_prevalence, $\lambda = 1.1$)	0.770 ± 0.019	0.707 ± 0.016	0.699 ± 0.018
patch_feature_oko (native_prevalence, $\lambda = 1.3$)	0.759 ± 0.011	0.683 ± 0.005	0.678 ± 0.003
patch_feature_progan_aug (native_prevalence, $\lambda = 0.8$)	0.762 ± 0.009	0.723 ± 0.004	0.692 ± 0.010
patch_feature_progan_aug (native_prevalence, $\lambda = 1.1$)	0.765 ± 0.013	0.698 ± 0.010	0.690 ± 0.011
patch_feature_progan_aug (native_prevalence, $\lambda = 1.3$)	0.751 ± 0.010	0.677 ± 0.007	0.686 ± 0.010
patch_feature_weighted_ce (native_prevalence, $\lambda = 0.8$)	0.771 ± 0.010	0.714 ± 0.004	0.698 ± 0.006
patch_feature_weighted_ce (native_prevalence, $\lambda = 1.1$)	0.768 ± 0.016	0.698 ± 0.014	0.690 ± 0.016
patch_feature_weighted_ce (native_prevalence, $\lambda = 1.3$)	0.759 ± 0.014	0.679 ± 0.007	0.677 ± 0.013

Table 3: Patch-feature test results for validation-selected configurations under constructed training distributions.

balanced accuracy (0.688), followed by OKO (0.683), CFAL (0.681), and weighted CE (0.679); CE+soft-MCC is the weakest patch method at 0.668. Declines from $\lambda = 0.8$ to $\lambda = 1.3$ range from three percentage points (CE+soft-F1: 0.718→0.688) to five (focal loss and ProGAN augmentation: 0.718→0.672 and 0.723→0.677). The total variation within the patch family is modest—approximately two percentage points across methods at any fixed λ—so the severity-induced decline is comparable to or larger than the spread between methods.

The WSI-bag regime shows higher absolute performance than the patch regime and a more uniform spread across methods (Table 4). MDE-MIL and RankMix are tied for the highest balanced accuracy at $\lambda = 0.8$ and $\lambda = 1.3$ (0.824 and 0.804 respectively). RankMix is the most stable method: its balanced accuracy holds at 0.824 from $\lambda = 0.8$ to $\lambda = 1.1$ and declines only to 0.804 at $\lambda = 1.3$—a two-point drop over the full range. MDE-MIL declines slightly more: 0.824→0.811→0.804. SC-MIL follows at 0.812–0.801–0.782. Among the loss-based variants, weighted CE is the most stable (0.802–0.794–0.789); focal MIL is nearly flat across severities (0.794–0.800–0.786), showing a non-monotone dip at the mildest imbalance. Balanced-sampling MIL occupies the lowest tier at each severity (0.809–0.791–0.777). The ranking of bag-composition methods (RankMix, MDE-MIL, SC-MIL) over loss-based variants is broadly maintained across all three severities, confirming that WSI-specific aggregation mechanisms preserve their advantage regardless of imbalance level.

Probability Calibration. In the patch regime, CFAL is a severe outlier: its affinity-based raw outputs are nearly flat over the 32 classes, with raw NLL of 3.2–3.3 and ECE of approximately 0.73 at all three λ values and a fitted temperature of $T = 0.05$. Post-hoc scaling concentrates rather than softens the distribution; after temperature scaling, CFAL's ECE is 0.401 at $\lambda = 0.8$ (remaining substantially miscalibrated) and 0.05–0.06 at $\lambda = 1.1$ and $\lambda = 1.3$. The remaining patch methods span a wide temperature range: OKO is under-confident before scaling ($T \approx 0.7$), focal loss and weighted CE are moderately over-confident ($T \approx 1.8$ and 2.5 respectively), and ProGAN augmentation requires strong concentration ($T \approx 4.2$); after

Method	Accuracy	Balanced accuracy	Macro F1
mde_mil (native_prevalence, $\lambda = 0.8$)	0.858 ± 0.010	0.824 ± 0.014	0.809 ± 0.006
mde_mil (native_prevalence, $\lambda = 1.1$)	0.852 ± 0.007	0.811 ± 0.004	0.808 ± 0.007
mde_mil (native_prevalence, $\lambda = 1.3$)	0.845 ± 0.008	0.804 ± 0.006	0.802 ± 0.016
mil_balanced_sampler_ce (native_prevalence, $\lambda = 0.8$)	0.837 ± 0.007	0.809 ± 0.005	0.782 ± 0.004
mil_balanced_sampler_ce (native_prevalence, $\lambda = 1.1$)	0.829 ± 0.012	0.791 ± 0.010	0.778 ± 0.020
mil_balanced_sampler_ce (native_prevalence, $\lambda = 1.3$)	0.817 ± 0.025	0.777 ± 0.013	0.768 ± 0.017
mil_focal (native_prevalence, $\lambda = 0.8$)	0.833 ± 0.005	0.794 ± 0.004	0.778 ± 0.006
mil_focal (native_prevalence, $\lambda = 1.1$)	0.843 ± 0.012	0.800 ± 0.004	0.795 ± 0.008
mil_focal (native_prevalence, $\lambda = 1.3$)	0.832 ± 0.006	0.786 ± 0.006	0.785 ± 0.003
mil_weighted_ce (native_prevalence, $\lambda = 0.8$)	0.834 ± 0.004	0.802 ± 0.006	0.778 ± 0.003
mil_weighted_ce (native_prevalence, $\lambda = 1.1$)	0.837 ± 0.009	0.794 ± 0.005	0.785 ± 0.001
mil_weighted_ce (native_prevalence, $\lambda = 1.3$)	0.833 ± 0.012	0.789 ± 0.009	0.785 ± 0.012
rankmix_mil (native_prevalence, $\lambda = 0.8$)	0.849 ± 0.002	0.824 ± 0.018	0.799 ± 0.018
rankmix_mil (native_prevalence, $\lambda = 1.1$)	0.851 ± 0.020	0.824 ± 0.017	0.804 ± 0.011
rankmix_mil (native_prevalence, $\lambda = 1.3$)	0.837 ± 0.014	0.804 ± 0.008	0.790 ± 0.009
sc_mil (native_prevalence, $\lambda = 0.8$)	0.845 ± 0.007	0.812 ± 0.021	0.794 ± 0.015
sc_mil (native_prevalence, $\lambda = 1.1$)	0.841 ± 0.012	0.801 ± 0.013	0.789 ± 0.015
sc_mil (native_prevalence, $\lambda = 1.3$)	0.832 ± 0.009	0.782 ± 0.016	0.783 ± 0.011

Table 4: WSI-bag test results for validation-selected configurations under constructed training distributions.

scaling, ECE falls below 0.04 for all non-CFAL methods. In the WSI-bag regime, RankMix is already well calibrated without scaling ($T \approx 1.1$, ECE ≈ 0.02), and MDE-MIL requires only minor adjustment ($T \approx 0.74$–0.83, ECE ≈ 0.05 after scaling); both achieve lower raw ECE than the loss-reweighting baselines ($T = 1.4$–2.2). After temperature scaling, all WSI-bag methods achieve ECE at or below 0.05. Calibration rankings are stable across imbalance severities, mirroring the discrimination pattern.

6 Discussion

The controlled construction reveals two complementary patterns. First, method rankings are broadly preserved across imbalance severities, with moderate reshuffling among closely performing methods. Second, balanced accuracy declines monotonically with λ for almost all methods, with a total range of three to five percentage points from the mildest to the strongest constructed regime. These patterns suggest that the relative utility of mitigation methods is largely robust to the particular severity chosen within this range, while absolute performance is sensitive to the support available for rare classes.

The monotone decline is similar across patch methods, ranging from three percentage points for CE+soft-F1 to five for focal loss and ProGAN augmentation. In contrast to earlier runs with the redistribution fallback, the within-method spread is tight—approximately two percentage points separating the best from the weakest patch method at any fixed λ—so the severity-induced decline is comparable to or larger than the advantage of choosing the best method. CE+soft-F1 shows the smallest decline and strongest relative performance at $\lambda = 1.3$ (0.688), while CE+soft-MCC is the weakest patch method throughout (0.668 at $\lambda = 1.3$).

In the WSI-bag regime, RankMix stands out for its stability: less than two percentage points of decline from $\lambda = 0.8$ to $\lambda = 1.3$, with performance unchanged from $\lambda = 0.8$ to $\lambda = 1.1$. Focal MIL is nearly flat across severities, showing a non-monotone response. MDE-MIL and SC-MIL decline somewhat more, while balanced-sampling MIL occupies the lowest tier at every severity. The ranking of bag-composition methods over loss-based variants is maintained throughout, indicating that the advantage of WSI-specific aggregation mechanisms is not contingent on the severity of training imbalance.

Method	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 0.8$)	1.339 ± 0.059	0.841 ± 0.023	0.386 ± 0.019	0.335 ± 0.013	0.156 ± 0.011	0.010 ± 0.002	2.441 ± 0.095
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 1.1$)	1.371 ± 0.068	0.869 ± 0.038	0.386 ± 0.029	0.338 ± 0.022	0.157 ± 0.012	0.016 ± 0.006	2.389 ± 0.140
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 1.3$)	1.456 ± 0.084	0.913 ± 0.037	0.411 ± 0.022	0.354 ± 0.013	0.166 ± 0.013	0.021 ± 0.001	2.467 ± 0.000
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 0.8$)	1.868 ± 0.051	0.877 ± 0.009	0.414 ± 0.014	0.336 ± 0.009	0.185 ± 0.007	0.015 ± 0.004	3.334 ± 0.195
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 1.1$)	1.783 ± 0.051	0.906 ± 0.027	0.419 ± 0.025	0.349 ± 0.016	0.183 ± 0.012	0.018 ± 0.002	3.116 ± 0.104
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 1.3$)	2.150 ± 0.184	0.933 ± 0.052	0.439 ± 0.030	0.352 ± 0.021	0.199 ± 0.014	0.016 ± 0.003	3.763 ± 0.072
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 0.8$)	1.524 ± 0.070	0.856 ± 0.036	0.404 ± 0.015	0.340 ± 0.012	0.171 ± 0.005	0.011 ± 0.003	2.789 ± 0.108
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 1.1$)	1.487 ± 0.068	0.888 ± 0.041	0.397 ± 0.023	0.344 ± 0.020	0.164 ± 0.010	0.017 ± 0.007	2.581 ± 0.132
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 1.3$)	1.490 ± 0.054	0.946 ± 0.041	0.421 ± 0.016	0.367 ± 0.013	0.169 ± 0.006	0.022 ± 0.005	2.495 ± 0.048
patch_feature_cfal (native_prevalence, $\lambda = 0.8$)	3.338 ± 0.002	1.394 ± 0.032	0.960 ± 0.000	0.506 ± 0.011	0.738 ± 0.011	0.401 ± 0.009	0.050 ± 0.000
patch_feature_cfal (native_prevalence, $\lambda = 1.1$)	3.212 ± 0.010	0.944 ± 0.063	0.950 ± 0.001	0.333 ± 0.022	0.734 ± 0.016	0.055 ± 0.006	0.050 ± 0.000
patch_feature_cfal (native_prevalence, $\lambda = 1.3$)	3.219 ± 0.002	0.993 ± 0.047	0.951 ± 0.000	0.348 ± 0.014	0.720 ± 0.013	0.053 ± 0.005	0.050 ± 0.000
patch_feature_focal (native_prevalence, $\lambda = 0.8$)	0.977 ± 0.040	0.793 ± 0.021	0.350 ± 0.015	0.323 ± 0.011	0.113 ± 0.011	0.009 ± 0.002	1.749 ± 0.068
patch_feature_focal (native_prevalence, $\lambda = 1.1$)	0.984 ± 0.032	0.805 ± 0.033	0.348 ± 0.023	0.323 ± 0.019	0.110 ± 0.010	0.013 ± 0.002	1.750 ± 0.088
patch_feature_focal (native_prevalence, $\lambda = 1.3$)	1.084 ± 0.081	0.865 ± 0.028	0.373 ± 0.017	0.343 ± 0.010	0.122 ± 0.016	0.018 ± 0.003	1.831 ± 0.122
patch_feature_oko (native_prevalence, $\lambda = 0.8$)	1.038 ± 0.066	0.885 ± 0.066	0.369 ± 0.024	0.332 ± 0.024	0.172 ± 0.007	0.039 ± 0.004	0.651 ± 0.022
patch_feature_oko (native_prevalence, $\lambda = 1.1$)	0.968 ± 0.039	0.872 ± 0.041	0.350 ± 0.019	0.328 ± 0.022	0.126 ± 0.017	0.035 ± 0.003	0.711 ± 0.028
patch_feature_oko (native_prevalence, $\lambda = 1.3$)	1.080 ± 0.032	0.922 ± 0.054	0.379 ± 0.008	0.341 ± 0.015	0.175 ± 0.017	0.033 ± 0.007	0.665 ± 0.013
patch_feature_progan_aug (native_prevalence, $\lambda = 0.8$)	2.341 ± 0.128	0.862 ± 0.032	0.422 ± 0.016	0.334 ± 0.012	0.195 ± 0.008	0.024 ± 0.004	4.205 ± 0.000
patch_feature_progan_aug (native_prevalence, $\lambda = 1.1$)	2.448 ± 0.142	0.879 ± 0.051	0.417 ± 0.021	0.331 ± 0.014	0.194 ± 0.009	0.028 ± 0.005	4.205 ± 0.000
patch_feature_progan_aug (native_prevalence, $\lambda = 1.3$)	2.556 ± 0.155	0.917 ± 0.038	0.443 ± 0.017	0.348 ± 0.013	0.206 ± 0.008	0.033 ± 0.001	4.300 ± 0.082
patch_feature_weighted_ce (native_prevalence, $\lambda = 0.8$)	1.311 ± 0.079	0.799 ± 0.030	0.376 ± 0.017	0.321 ± 0.014	0.155 ± 0.009	0.012 ± 0.002	2.524 ± 0.098
patch_feature_weighted_ce (native_prevalence, $\lambda = 1.1$)	1.282 ± 0.048	0.809 ± 0.034	0.374 ± 0.025	0.322 ± 0.019	0.150 ± 0.010	0.009 ± 0.000	2.468 ± 0.082
patch_feature_weighted_ce (native_prevalence, $\lambda = 1.3$)	1.323 ± 0.137	0.839 ± 0.047	0.389 ± 0.025	0.333 ± 0.016	0.157 ± 0.017	0.013 ± 0.005	2.468 ± 0.082

Table 5: Patch-feature calibration metrics for validation-selected configurations under constructed training distributions. NLL, Brier, and ECE are reported before (raw) and after (TS) post-hoc temperature scaling; T is the per-seed mean fitted temperature.

Method	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
mde_mil (native_prevalence, $\lambda = 0.8$)	0.614 ± 0.055	0.594 ± 0.059	0.223 ± 0.019	0.220 ± 0.020	0.042 ± 0.001	0.046 ± 0.012	0.831 ± 0.016
mde_mil (native_prevalence, $\lambda = 1.1$)	0.651 ± 0.021	0.623 ± 0.024	0.233 ± 0.007	0.228 ± 0.008	0.054 ± 0.006	0.051 ± 0.006	0.813 ± 0.032
mde_mil (native_prevalence, $\lambda = 1.3$)	0.714 ± 0.042	0.652 ± 0.059	0.251 ± 0.014	0.238 ± 0.016	0.091 ± 0.011	0.048 ± 0.007	0.735 ± 0.014
mil_balanced_sampler_ce (native_prevalence, $\lambda = 0.8$)	0.874 ± 0.041	0.599 ± 0.016	0.260 ± 0.011	0.237 ± 0.008	0.101 ± 0.006	0.019 ± 0.007	2.066 ± 0.040
mil_balanced_sampler_ce (native_prevalence, $\lambda = 1.1$)	1.002 ± 0.033	0.651 ± 0.036	0.283 ± 0.013	0.252 ± 0.011	0.113 ± 0.003	0.020 ± 0.003	2.208 ± 0.042
mil_balanced_sampler_ce (native_prevalence, $\lambda = 1.3$)	0.973 ± 0.102	0.690 ± 0.099	0.299 ± 0.042	0.269 ± 0.038	0.110 ± 0.012	0.022 ± 0.008	2.091 ± 0.119
mil_focal (native_prevalence, $\lambda = 0.8$)	0.670 ± 0.015	0.571 ± 0.009	0.247 ± 0.005	0.233 ± 0.005	0.080 ± 0.005	0.021 ± 0.002	1.548 ± 0.052
mil_focal (native_prevalence, $\lambda = 1.1$)	0.628 ± 0.040	0.574 ± 0.035	0.237 ± 0.015	0.231 ± 0.013	0.052 ± 0.005	0.015 ± 0.000	1.401 ± 0.047
mil_focal (native_prevalence, $\lambda = 1.3$)	0.643 ± 0.047	0.591 ± 0.037	0.248 ± 0.015	0.240 ± 0.014	0.060 ± 0.002	0.017 ± 0.001	1.385 ± 0.026
mil_weighted_ce (native_prevalence, $\lambda = 0.8$)	0.859 ± 0.039	0.575 ± 0.013	0.258 ± 0.010	0.231 ± 0.004	0.107 ± 0.007	0.023 ± 0.004	2.137 ± 0.108
mil_weighted_ce (native_prevalence, $\lambda = 1.1$)	0.897 ± 0.042	0.599 ± 0.035	0.266 ± 0.013	0.238 ± 0.013	0.106 ± 0.003	0.022 ± 0.005	2.184 ± 0.042
mil_weighted_ce (native_prevalence, $\lambda = 1.3$)	0.841 ± 0.090	0.593 ± 0.047	0.269 ± 0.025	0.240 ± 0.020	0.105 ± 0.012	0.018 ± 0.003	2.043 ± 0.040
rankmix_mil (native_prevalence, $\lambda = 0.8$)	0.552 ± 0.011	0.549 ± 0.010	0.224 ± 0.003	0.224 ± 0.002	0.024 ± 0.007	0.017 ± 0.001	1.062 ± 0.054
rankmix_mil (native_prevalence, $\lambda = 1.1$)	0.574 ± 0.052	0.572 ± 0.052	0.223 ± 0.025	0.224 ± 0.024	0.014 ± 0.004	0.018 ± 0.009	1.049 ± 0.020
rankmix_mil (native_prevalence, $\lambda = 1.3$)	0.584 ± 0.040	0.584 ± 0.040	0.235 ± 0.020	0.235 ± 0.020	0.024 ± 0.001	0.021 ± 0.006	1.050 ± 0.054
sc_mil (native_prevalence, $\lambda = 0.8$)	0.755 ± 0.114	0.550 ± 0.053	0.245 ± 0.019	0.224 ± 0.018	0.091 ± 0.009	0.018 ± 0.010	1.977 ± 0.101
sc_mil (native_prevalence, $\lambda = 1.1$)	0.790 ± 0.093	0.569 ± 0.046	0.252 ± 0.022	0.230 ± 0.021	0.094 ± 0.010	0.019 ± 0.010	1.979 ± 0.135
sc_mil (native_prevalence, $\lambda = 1.3$)	0.792 ± 0.085	0.577 ± 0.046	0.263 ± 0.019	0.237 ± 0.015	0.100 ± 0.010	0.022 ± 0.004	2.021 ± 0.067

Table 6: WSI-bag calibration metrics for validation-selected configurations under constructed training distributions. Columns as in Table 5.

Calibration analysis complements discrimination: CFAL leads on patch discrimination but requires temperature scaling to produce reliable probability outputs, while MDE-MIL and RankMix improve both discrimination and raw calibration in the WSI-bag regime, though the gap relative to loss-based baselines narrows after a shared post-hoc scaling step.

Taken together, these results extend the companion benchmark in a controlled direction. The companion demonstrated that class frequency alone does not predict classwise difficulty in the native TCGA-UT distribution. The present experiment shows that when frequency is imposed directly via a clean power-law construction, aggregate discrimination declines predictably but modestly with imbalance severity, and the method ranking remains informative across the range tested. The prediction unit—patch versus WSI-bag—continues to be the dominant determinant of absolute performance: WSI-bag methods achieve balanced accuracy approximately ten percentage points higher than their patch-feature counterparts at equivalent imbalance.

7 Conclusion

This report defines a full-scale controlled TCGA-UT imbalance benchmark that keeps all 32 cancer types, preserves patient-disjoint validation and test splits, and varies the training support profile under native-prevalence class ordering at imbalance parameters $\lambda \in \{0.8, 1.1, 1.3\}$. A fixed pool of 1,528 slides ensures that the power-law construction remains capping-free at all three severities, yielding realized head-to-tail ratios of approximately 15:1, 43:1, and 78:1 that bracket the native TCGA-UT long tail. The primary finding across all three constructed regimes is that method rankings are broadly preserved while balanced accuracy declines monotonically by three to five percentage points from $\lambda = 0.8$ to $\lambda = 1.3$. At the patch level, CFAL, OKO, and ProGAN augmentation lead at mild imbalance (0.723), while CE+soft-F1 achieves the highest balanced accuracy at the strongest regime (0.688); CE+soft-MCC is the weakest method throughout. The within-method spread is narrow, so severity-induced decline is comparable to or larger than method choice effects. At the WSI-bag level, MDE-MIL and RankMix lead across all severities; RankMix is the most stable method, with a two-point decline over the full range. The broadly preserved ranking across severities indicates that method selection can be informed by experiments at mild imbalance, while the monotone decline confirms that stronger imbalance meaningfully degrades absolute performance. The prediction unit remains the dominant determinant: WSI-bag methods achieve balanced accuracy roughly ten percentage points above patch-feature methods at equivalent imbalance severity. Temperature scaling concentrates CFAL's near-flat affinity-based outputs and narrows the raw calibration advantage of MDE-MIL and RankMix, confirming that the best discriminator and the best-calibrated method are not always the same, and that a post-hoc step is sufficient to align probability outputs across all methods evaluated.

References

Glenn W. Brier. Verification of forecasts expressed in terms of probabilities. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078⟨0001:VOFEIT⟩2.0.CO;2.

Yuan-Chih Chen and Chun-Shien Lu. RankMix: Data augmentation for weakly supervised learning of classifying whole slide images with diverse sizes and imbalanced categories. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23936–23945, 2023. doi: 10.1109/CVPR52729.2023.02397.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR, 2017.

Dinkar Juyal, Siddhant Shingi, Syed Ashar Javed, Harshith Padigela, Chintan Shah, Anand Sampat, Archit Khosla, John Abel, and Amaro Taylor-Weiner. SC-MIL: Supervised contrastive multiple instance learning for imbalanced classification in pathology. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 7016–7025, 2024. doi: 10.1109/WACV57761.2024.00702.

Daisuke Komura, Akihiro Kawabe, Kenta Fukuta, Kazuya Sano, Tsukasa Umezaki, Haruka Koda, Riku Suzuki, Keisuke Tominaga, Michiaki Ochi, Hirofumi Konishi, et al. Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424, 2022. doi: 10.1016/j.celrep.2022.110424.

Xitong Ling, Yifeng Ping, Jiawen Li, Jing Peng, and Yuxuan Chen. MDE-MIL. *Dos mil tres mil*, 27, 2025. doi: 10.35707/dostresmil/27.

Mahdi Pakdaman Naeini, Gregory Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 29, pages 2901–2907, 2015. doi: 10.1609/aaai.v29i1.9602.

Daniel Scholz, Ayhan Can Erdur, Josef Buchner, Jan C. Peeken, Daniel Rueckert, and Benedikt Wiestler. Imbalance-aware loss functions improve medical image classification. In *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning*, volume 250 of *Proceedings of Machine Learning Research*, pages 1341–1356. PMLR, 2024.

Katarzyna Tomczak, Patrycja Czerwinska, and Maciej Wiznerowicz. The cancer genome atlas (TCGA): an immeasurable source of knowledge. *Contemporary Oncology*, 19(1A):A68–A77, 2015. doi: 10.5114/wo.2014.47136.

Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, Eric Zimmermann, et al. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935, 2024. doi: 10.1038/s41591-024-03141-0.

Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, and George Shaikovski. Virchow2. *2024 Shell TechXplorer Digest*, 2024(1), 2024. doi: 10.52196/208316.24.1.